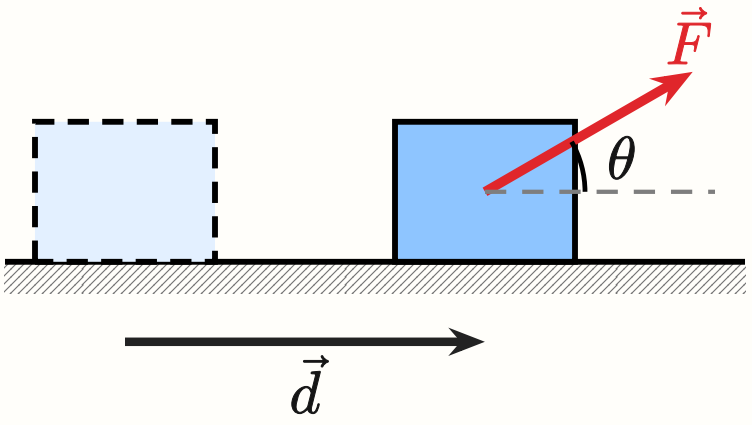
Work by a Constant Force

Work, Energy and Power · Class 11

1 Definition



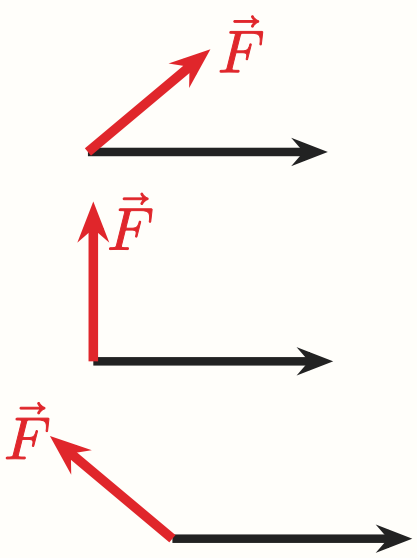
$$W = \vec{F} \cdot \vec{d} = Fd\cos\theta$$

$\vec{F}$ constant · $\vec{d}$: displacement of the point where $\vec{F}$ acts · SCALAR · $1\,\mathrm{J} = 1\,\mathrm{N\,m}$, $[\mathrm{ML^2T^{-2}}]$

Example

$F = 10\,\mathrm{N}$ at 60° to a level floor, block moved $5\,\mathrm{m}$ along the floor: $W = 10 \times 5 \times \cos 60^\circ = 25\,\mathrm{J}$.

2 Sign of work



$\theta < 90^\circ$: gravity, falling ball — $W > 0$

$\theta = 90^\circ$: N on a fixed floor — $W = 0$

$\theta > 90^\circ$: kinetic friction — $W < 0$

Golden points

- ★ Zero work: $d = 0$ (pushing a fixed wall) or $\vec{F} \perp \vec{d}$ (centripetal force; load carried at constant height).
- ★ Work depends on the FRAME (displacement does).

Work by Common Forces

1 Gravity

$$W_g = mg\,(h_i - h_f)$$

g uniform (near the surface) · ANY path: only the drop in height matters

Example

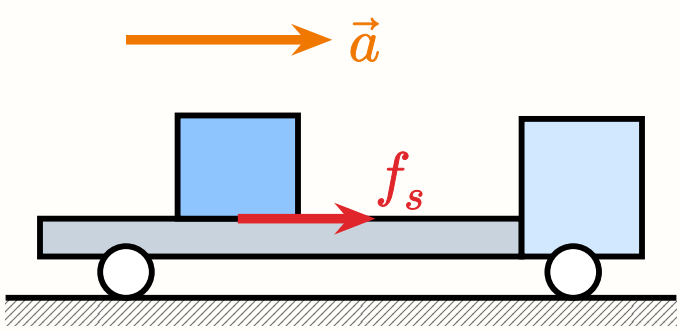
$2\,\mathrm{kg}$ *lifted slowly through* $3\,\mathrm{m}$: $W_g = -60\,\mathrm{J}$, $W_{hand} = +60\,\mathrm{J}$, $W_{net} = 0$ *(speed unchanged).*

2 Normal force

- *FIXED surface:* $\vec{N} \perp \vec{v}$, *so* $W_N = 0$.
- *MOVING surface:* $W_N \neq 0$ *(person in a rising lift: positive).*

  Block on a sliding smooth wedge: N *does work on each, total zero.*

3 Friction



- *Sliding on a FIXED surface:* $W = -\mu_k N s$.
- *Static friction CAN do positive work (box on an accelerating truck).*

Golden points

★ *A kinetic-friction pair: total work* $= -f_k \times$*(relative sliding distance)* < 0*: heat. A static pair: total work* 0.

★ *Ideal string: tensions on the two bodies do equal and opposite work (total* 0*).*

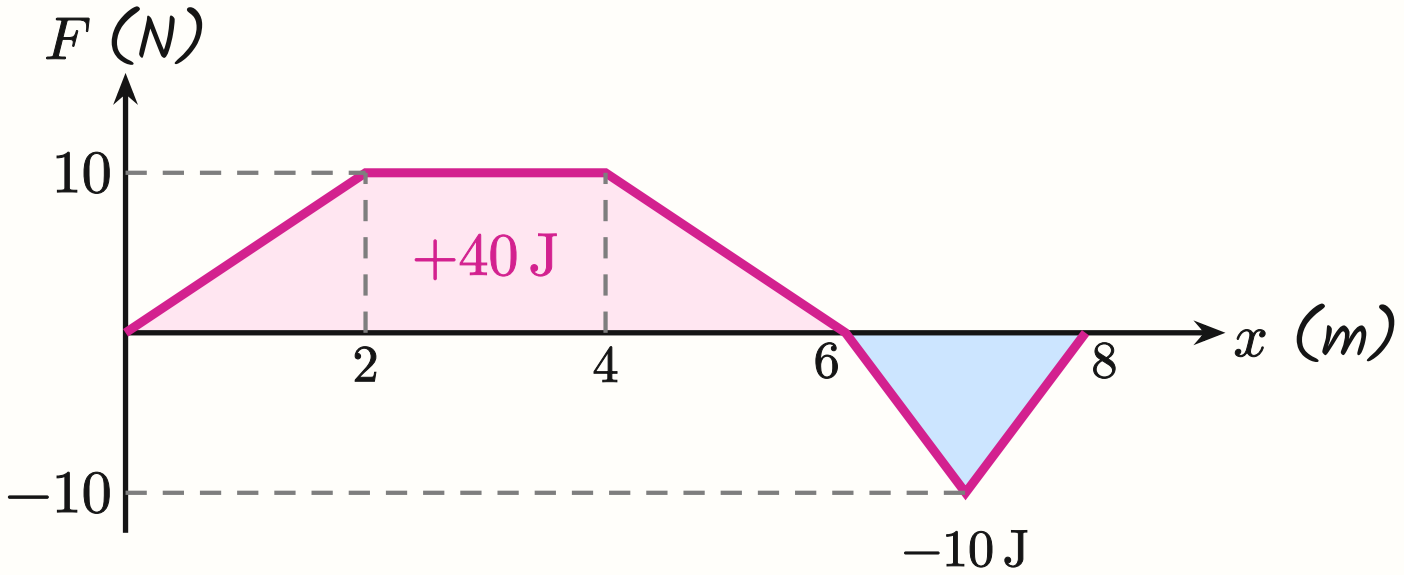
Work by a Variable Force

1 Integral form

$$W = \int_{x_1}^{x_2} F(x)\,dx$$

force along the line of motion, varying with position · sum of tiny steps $F\,dx$

2 Area under the F–x graph



- *Area ABOVE the axis: positive work; BELOW: negative.* 0–6 m: $\frac{1}{2}(2)(10) + (2)(10) + \frac{1}{2}(2)(10) = 40\,\text{J}$; 6–8 m: $-10\,\text{J}$; *net* $W = 30\,\text{J}$.
- $1\,\text{kg}$, *this force alone, from rest at* $x = 0$: $K = 30\,\text{J}$ *at* $x = 8\,\text{m}$, $v = \sqrt{60} \approx 7.75\,\text{m s}^{-1}$.

Example

$F = 3x^2\,\text{N}$ (x *in m*) *from* $x = 0$ *to* $x = 2\,\text{m}$: $W = \left[x^3\right]_0^2 = 8\,\text{J}$.

Golden points

★ F *given as a function of* t *or* v: *use* $W = \Delta K$ *or* $W = \int P\,dt$.

Work Along a Path

1 Line integral

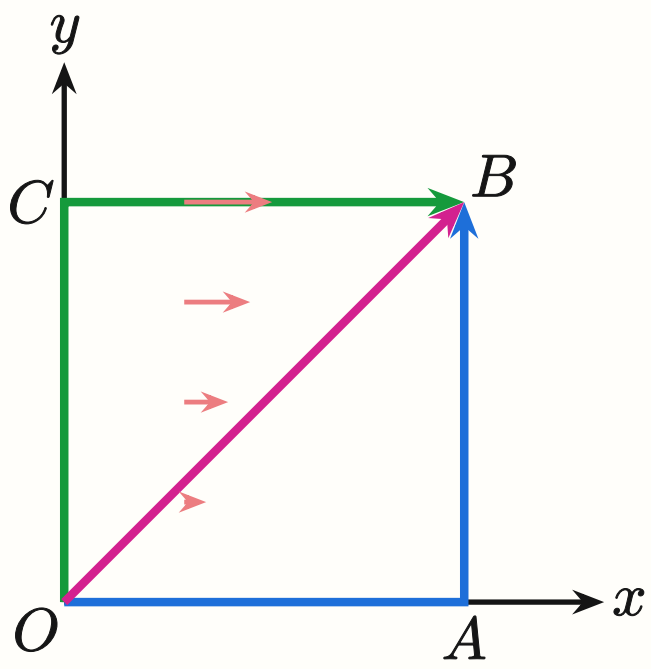
$$W_{A\to B} = \int_A^B \vec{F}\cdot d\vec{r} = \int (F_x\,dx + F_y\,dy + F_z\,dz)$$

$\vec{F}$ may vary along the path · use the ACTUAL path (unless the force is conservative)

Example

- *Constant* $\vec{F} = (3\hat{i} + 4\hat{j})\,\text{N}$, $\vec{d} = (2\hat{i} - \hat{j} + 3\hat{k})\,\text{m}$:
  $W = 6 - 4 + 0 = 2\,\text{J}$.
- $\vec{F} = (2x\,\hat{i} + 3y^2\hat{j})\,\text{N}$, $(0,0) \to (1\,\text{m}, 1\,\text{m})$:
  $W = \int_0^1 2x\,dx + \int_0^1 3y^2 dy = 2\,\text{J}$ *on ANY path.*

2 Path matters for $\vec{F} = y\,\hat{i}$ (N)



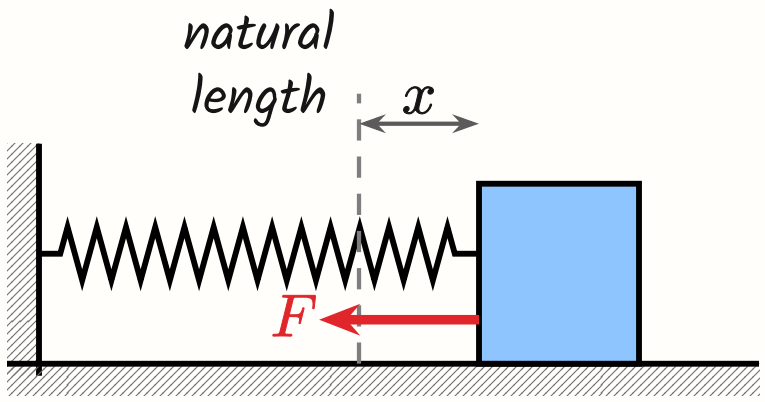
- $O \to A \to B$: $F = 0$ *on* OA, $\vec{F} \perp d\vec{r}$ *on* AB: $W = 0$.
- $O \to C \to B$: *on* CB $y = 1$: $W = 1\,\text{J}$.
- *Straight* OB $(y = x)$: $W = \int_0^1 x\,dx = \frac{1}{2}\,\text{J}$.

Golden points

★ *Path-dependent work:* $\vec{F} = y\,\hat{i}$ *is NON-conservative (no* U *exists).*

Work Done by a Spring

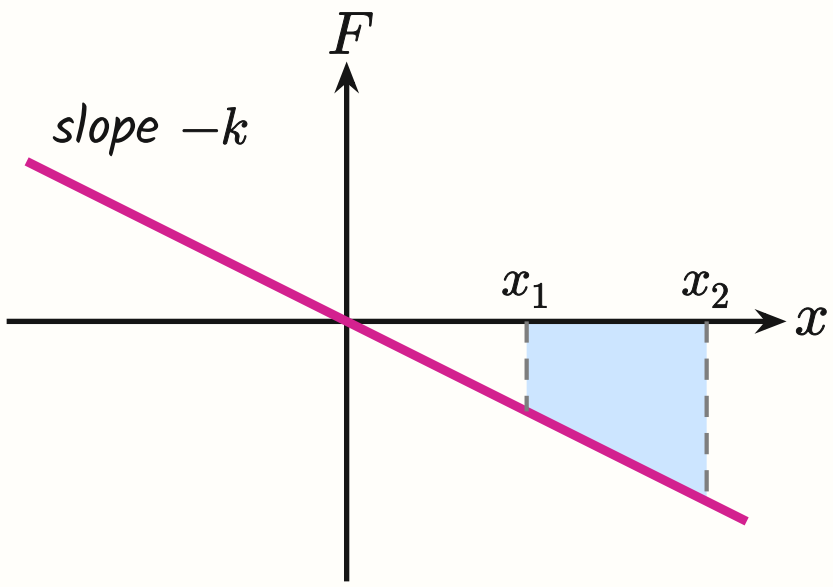
1 Spring force



$$F = -kx$$

x: extension (+) or compression (−) from the natural length · k: spring constant ($\mathrm{N\,m^{-1}}$) · ideal massless spring, elastic limit

2 Work by the spring



$$W_s = -\tfrac{1}{2}k\left(x_2^2 - x_1^2\right)$$

from deformation x_1 to x_2 · any path, any speed · $= -$(shaded area)

Example

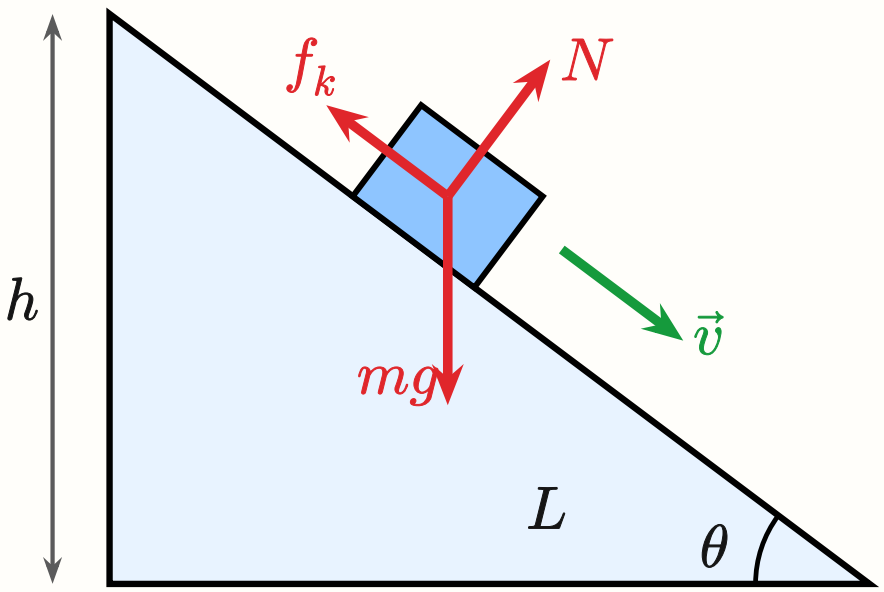
$k = 200\,\mathrm{N\,m^{-1}}$: stretched $0 \to 10\,\mathrm{cm}$: $W_s = -1\,\mathrm{J}$; then $10 \to 20\,\mathrm{cm}$: $W_s = -3\,\mathrm{J}$ (NOT $-1\,\mathrm{J}$); back $20 \to 10\,\mathrm{cm}$: $W_s = +3\,\mathrm{J}$.

Golden points

- ★ Depends only on the initial and final deformation (conservative force).
- ★ An agent stretching it SLOWLY ($\Delta K = 0$) from natural length does $W_{agent} = -W_s = +\tfrac{1}{2}kx^2$.

Gravity and Friction on an Incline

1 Block sliding down a rough fixed incline



$$W_g = mgh = mgL\sin\theta$$

$$W_f = -\mu_k mg\cos\theta\, L$$

$W_N = 0$ (incline fixed) · f_k up the slope, opposite to the sliding

Example

$m = 2\,\text{kg}$, $\theta = 37^\circ$ ($\sin = 0.6$, $\cos = 0.8$), $L = 5\,\text{m}$, $\mu_k = 0.5$:
$W_g = +60\,\text{J}$, $W_f = -40\,\text{J}$, $W_N = 0$, $W_{net} = 20\,\text{J}$. From rest:
$v_{bottom} = \sqrt{20} \approx 4.47\,\text{m s}^{-1}$.

2 Pulled up slowly along the incline

- Force parallel to the slope, $\Delta K = 0$:
  $W_F = mgh + \mu_k mg\cos\theta\, L = 60 + 40 = 100\,\text{J}$ (same block).
- Smooth incline ($\mu = 0$): $W_F = mgh$, the same as lifting straight up.

Golden points

★ Gravity: only the height matters. Friction: the PATH LENGTH matters; on a rough level floor $W_f = -\mu_k mg \times$(total distance), even for a back-and-forth trip.

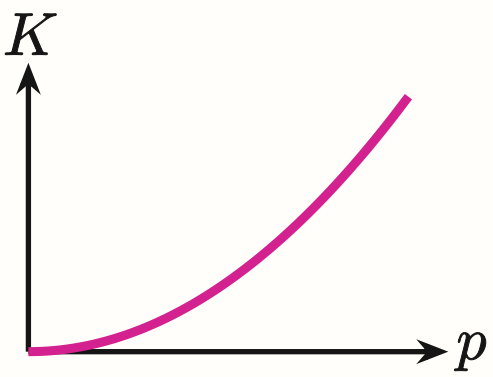
Kinetic Energy and Momentum

1 Kinetic energy

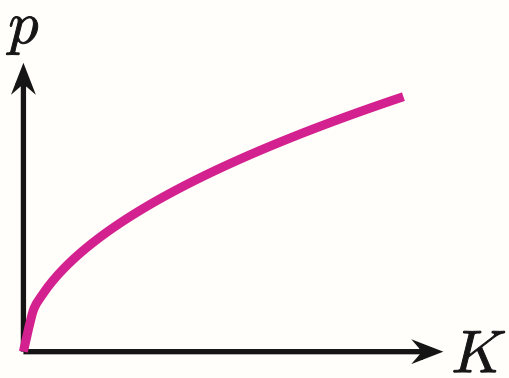
$$K = \tfrac{1}{2}mv^2 = \frac{p^2}{2m}, \qquad p = \sqrt{2mK}$$

speed $v \ll c$ · scalar, never negative · depends on the frame

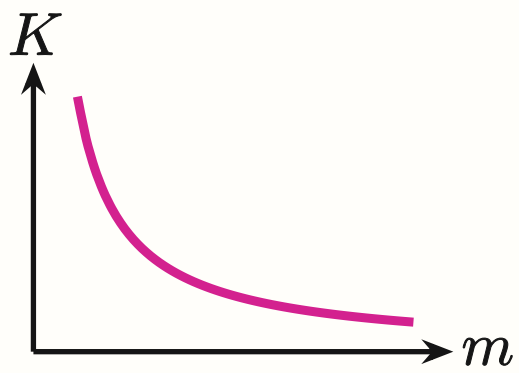
2 Graphs



m fixed: parabola



$p \propto \sqrt{K}$



same p: $K \propto 1/m$

3 Comparisons and percentage changes

- *Same p: LIGHTER body has more K ($2\,\mathrm{kg}$ vs $8\,\mathrm{kg}$; $4:1$). Same K: HEAVIER has more p.*
- *p up 20%: $K \times 1.44$, up 44%. K up 300% ($\times 4$): p doubles (up 100%). K doubled: p up 41.4%.*

Golden points

★ *K changes by WORK, p by IMPULSE: uniform circular motion changes $\vec{p}$, not K.*

★ *Total p can be zero while total K is not (equal and opposite momenta).*

Work-Energy Theorem

1 Statement

$$W_{net} = K_f - K_i = \Delta K$$

W_{net}: work by ALL forces on a particle · inertial frame · system: include internal work (p9)

$$F = m\frac{dv}{dt} = mv\frac{dv}{dx} \;\Rightarrow\; \int F\,dx = \tfrac{1}{2}mv_f^2 - \tfrac{1}{2}mv_i^2$$

proof for straight-line motion; the same works component by component for any path

Example

$2\,\mathrm{kg}$ speeded up from 2 to $6\,\mathrm{m\,s^{-1}}$: $W_{net} = \frac{1}{2}(2)(36-4) = 32\,\mathrm{J}$.

2 Stopping distance

$$d = \frac{u^2}{2\mu_k g} = \frac{K}{F}$$

sliding on a rough level floor (μ_k constant), or any constant retarding force F

Example

$u = 10\,\mathrm{m\,s^{-1}}$, $\mu_k = 0.5$: $d = 10\,\mathrm{m}$, whatever the mass. Same F: same K stops in the same distance; same p: $d = p^2/2mF$, the lighter body goes farther.

Golden points

★ Bullet halves its speed in one plank: it lost $\frac{3}{4}$ of K; the remaining $\frac{1}{4}$ stops it after $\frac{1}{3}$ plank more (same resistive force).

Systems and Non-inertial Frames

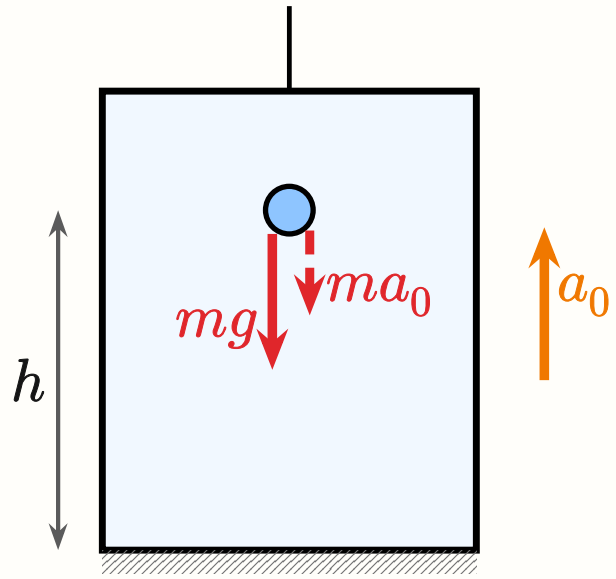
1 System of particles

$$W_{ext} + W_{int} = \Delta K_{system}$$

internal forces cancel in $\sum \vec{F}$, but their WORK need not

- *Explosion, blocks pushed apart by a spring: K rises, NO external work.*
- *Jumping: the floor's N does zero work (contact point at rest); K comes from muscles.*

2 Non-inertial frame



$$W_{real} + W_{pseudo} = \Delta K'$$

frame accelerating with $\vec{a}_0$: pseudo force $-m\vec{a}_0$ on each body · K' measured in that frame

Example

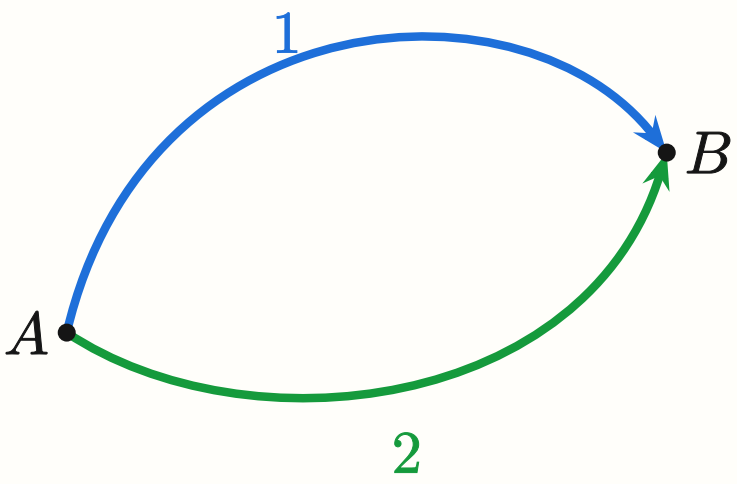
Lift accelerating UP at $a_0 = 2\,\mathrm{m\,s^{-2}}$, ball released $h = 1.5\,\mathrm{m}$ above the floor: $m(g + a_0)h = \frac{1}{2}mv'^2$, $v' = \sqrt{2(12)(1.5)} = 6\,\mathrm{m\,s^{-1}}$ relative to the floor.

Golden points

★ *Work and K have different values in different frames, but $W_{net} = \Delta K$ holds in EVERY inertial frame.*

Conservative and Non-conservative

1 Conservative force



- Work from A to B is the SAME on every path: $W_1 = W_2$.
- Equivalently: work round any CLOSED path is zero.
- A potential energy U exists: $W = -\Delta U$.

$$\oint \vec{F} \cdot d\vec{r} = 0$$

true for gravity, ideal spring force, electrostatic force, any CONSTANT force

2 Non-conservative force

- Work depends on the path; NOT zero round a loop: kinetic friction, air drag, viscous force. No U; mechanical energy becomes heat.

Example

$5\,\mathrm{kg}$ dragged $3\,\mathrm{m}$ out and $3\,\mathrm{m}$ back on a rough level floor ($\mu_k = 0.2$): $W_f = -0.2 \times 5 \times 10 \times 6 = -60\,\mathrm{J}$, while gravity does 0 round the trip.

Golden points

★ 2D test: conservative if $\partial F_x/\partial y = \partial F_y/\partial x$ everywhere (region without holes). $y\,\hat{i}$: $1 \neq 0$, not conservative; $y\,\hat{i} + x\,\hat{j}$: conservative, $U = -xy$.

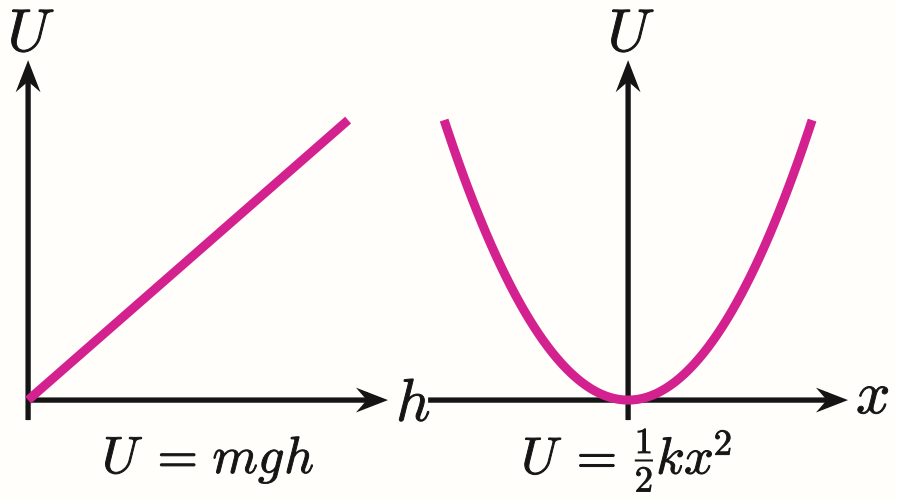
Potential Energy

1 Definition

$$\Delta U = U_B - U_A = -W_{cons} = -\int_A^B \vec{F} \cdot d\vec{r}$$

ONLY for a conservative force · the zero of U is our choice; only CHANGES of U are physical

2 Two standard forms



$$U = mgh$$

$h \ll R_E$ (g constant) · h above the chosen zero level (below it $U < 0$)

$$U = \tfrac{1}{2}kx^2$$

x from the NATURAL length · same for stretch and compression

Example

$2\,\text{kg}$ *raised* $3\,\text{m}$: $\Delta U = +60\,\text{J}$. *Spring* $400\,\text{N m}^{-1}$ *compressed* $5\,\text{cm}$: $U = 0.5\,\text{J}$; *to* $10\,\text{cm}$: $U = 2\,\text{J}$ *(extra* $1.5\,\text{J}$, *not* $0.5\,\text{J}$*).*

Golden points

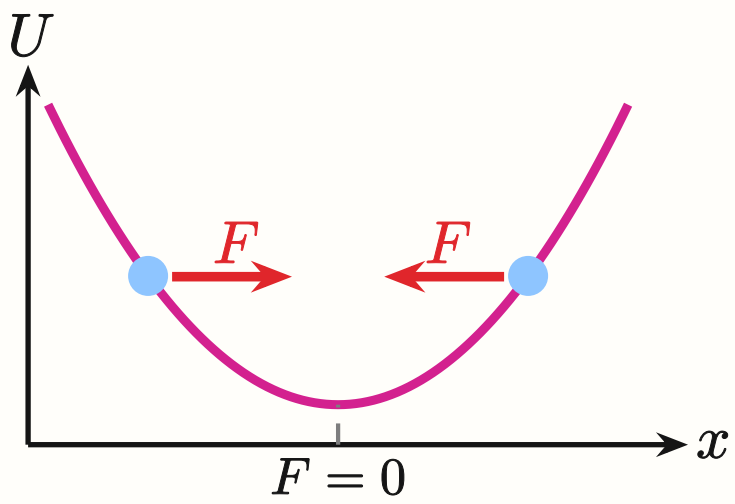
- ★ *General gravity:* $U = -GMm/r$ *(Gravitation chapter); it reduces to* mgh *for* $h \ll R_E$.
- ★ *Same force* F *on a spring:* $U = F^2/2k$ *(softer spring stores MORE). Same extension: stiffer spring stores more.*

Force from Potential Energy

1 Force is minus the slope

$$F_x = -\frac{dU}{dx}, \qquad \vec{F} = -\left(\frac{\partial U}{\partial x}\hat{i} + \frac{\partial U}{\partial y}\hat{j} + \frac{\partial U}{\partial z}\hat{k}\right)$$

conservative force · $\vec{F}$ points towards DECREASING U



- *U rising with x (slope $+$): F towards $-x$.*
- *U falling (slope $-$): F towards $+x$.*
- *Flat U (slope 0): $F = 0$.*

2 Examples

- *Spring: $U = \frac{1}{2}kx^2 \Rightarrow F = -kx$. Gravity: $U = mgy \Rightarrow F_y = -mg$.*
- *$U = (4x^2 - 3y)\,\mathrm{J}$ (x, y in m): $\vec{F} = (-8x\,\hat{i} + 3\,\hat{j})\,\mathrm{N}$; at $(1\,\mathrm{m}, 2\,\mathrm{m})$: $\vec{F} = (-8\hat{i} + 3\hat{j})\,\mathrm{N}$, $|\vec{F}| = \sqrt{73} \approx 8.54\,\mathrm{N}$.*

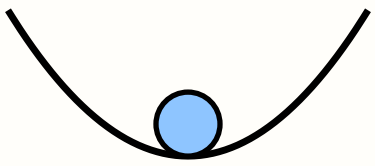
3 Back from force to energy

$$U(x) = -\int F\,dx + C$$

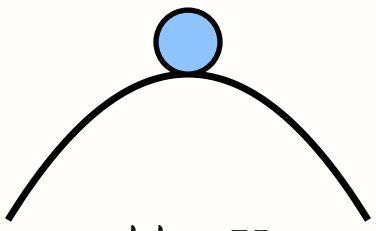
C fixed by the chosen zero · $F = -\alpha x^3$ gives $U = \frac{1}{4}\alpha x^4$ (zero at $x = 0$)

Equilibrium from $U(x)$

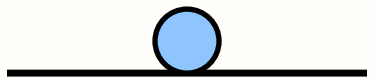
1 Three kinds



stable: U min



unstable: U max

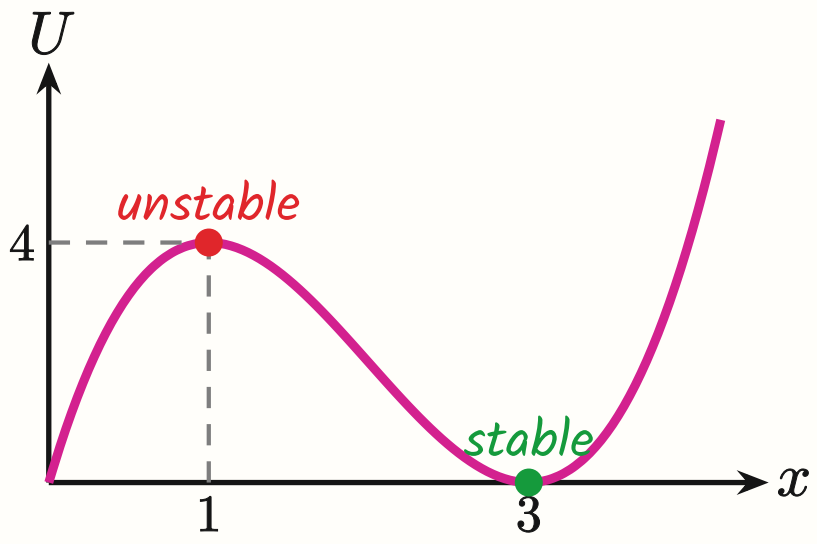


neutral: U flat

$$\frac{dU}{dx} = 0; \quad \frac{d^2U}{dx^2} > 0 \text{ stable}, \; < 0 \text{ unstable}$$

neutral: U constant over a range · stable: a small push gives a RESTORING force

2 Worked example



- $U = x^3 - 6x^2 + 9x$ (J, x in m).
- $U' = 3(x-1)(x-3) = 0$ at $x = 1$ and $3\,\mathrm{m}$.
- $U'' = 6x - 12$: -6 at $x = 1$ (unstable), $+6$ at $x = 3$ (stable).

Golden points

★ Small oscillations about a STABLE point x_0: $\omega = \sqrt{U''(x_0)/m}$ (small amplitude). Here $m = 2\,\mathrm{kg}$ at $x = 3\,\mathrm{m}$: $\omega = \sqrt{3} \approx 1.73\,\mathrm{rad\,s^{-1}}$.

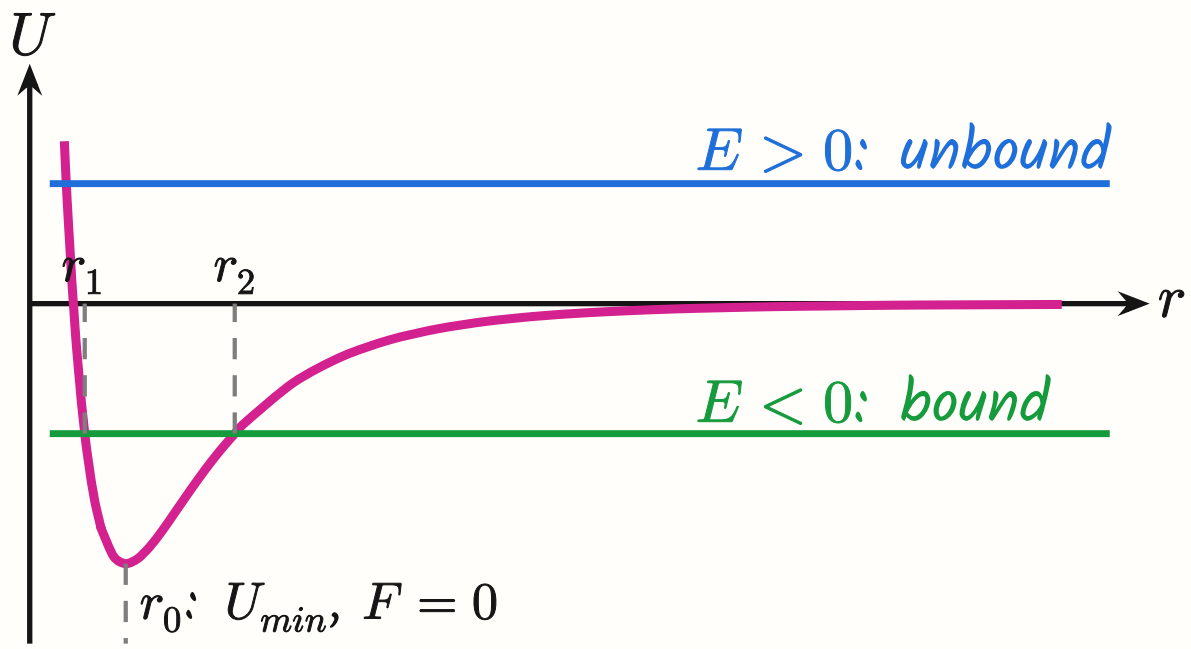
Reading a $U(x)$ Graph

1 Total energy line

$$K = E - U \geq 0$$

$E = K + U$ constant (only conservative forces) · motion only where $U \leq E$ · turning points ($v = 0$) where $U = E$

2 Two-atom (molecule) potential



- $E < 0$: the atoms oscillate between the turning points r_1 and r_2 (bound molecule).
- $E > 0$: no outer turning point, the atoms fly apart. Energy needed to separate from rest at r_0 (dissociation): $|U_{min}|$, with $U(\infty) = 0$.
- $r < r_0$: slope negative, $F > 0$ (repulsion); $r > r_0$: slope positive, $F < 0$ (attraction).

Golden points

★ From a U-x graph: F is zero at every maximum and minimum, and largest where the graph is steepest.

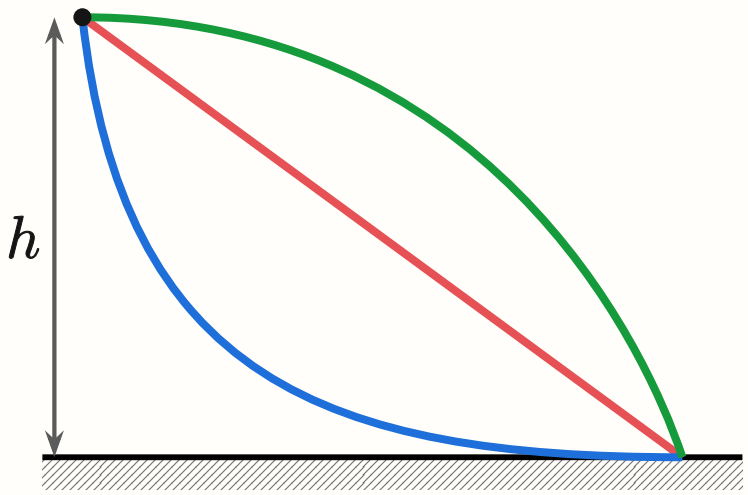
Conservation of Mechanical Energy

1 Statement

$$K_i + U_i = K_f + U_f$$

only CONSERVATIVE forces do work (others, like N on a fixed smooth surface, may act but do no work) · one zero level for U

2 Speed depends only on the drop



$v = \sqrt{2gh}$ *on each*

- *Smooth tracks, released from rest at height h: the same speed at the bottom on every track.*
- *The TIMES differ (they depend on the shape).*

Example

Ball dropped from $20\,\mathrm{m}$ (air drag neglected): $v = 20\,\mathrm{m\,s^{-1}}$ at the ground; $10\,\mathrm{m\,s^{-1}}$ after falling $5\,\mathrm{m}$; $K = U$ at $10\,\mathrm{m}$ height.

3 Simple pendulum

$$v_{low} = \sqrt{2gL(1 - \cos\theta_0)}$$

released from rest at angle θ_0 · light string, no drag (tension does no work: $\perp \vec{v}$) · $L = 2.5\,\mathrm{m}$, $\theta_0 = 60°$: $5\,\mathrm{m\,s^{-1}}$

Golden points

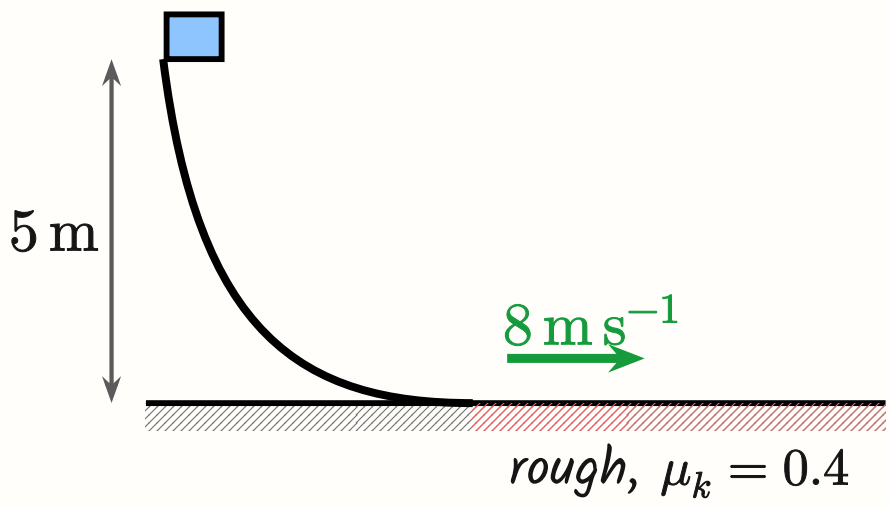
★ *Free fall from H: U falls and K rises linearly with the drop; $K + U = mgH$ at every height.*

Non-conservative Work

1 Mechanical energy changes

$$W_{nc} = \Delta K + \Delta U = \Delta E_{mech}$$

W_{nc}: work by friction, drag, applied forces · $W_{nc} < 0$: mechanical energy becomes heat, sound, ...



Example

2 kg from rest at 5 m down a rough curve; speed $8\,\mathrm{m\,s^{-1}}$ at the bottom:
$W_f = \frac{1}{2}(2)(64) - 2(10)(5) = 64 - 100 = -36\,\mathrm{J}$. Then on the rough floor it stops after $\dfrac{64}{0.4 \times 2 \times 10} = 8\,\mathrm{m}$.

2 Heat from kinetic friction

$$Q = f_k \times s_{rel}$$

s_{rel}: distance slid RELATIVE to the other surface (not relative to the ground)

Example

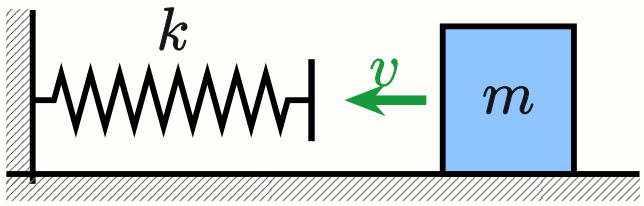
1 kg placed on a belt moving at a steady $2\,\mathrm{m\,s^{-1}}$, $\mu_k = 0.2$: it slides 1 m relative to the belt; heat $= 2\,\mathrm{J}$, its $K = 2\,\mathrm{J}$, the belt motor supplies $4\,\mathrm{J}$.

Golden points

★ TOTAL energy (mechanical + internal + heat + ...) of an isolated system is ALWAYS conserved; only mechanical energy is lost.

Spring–Block Problems

1 Block hits a spring



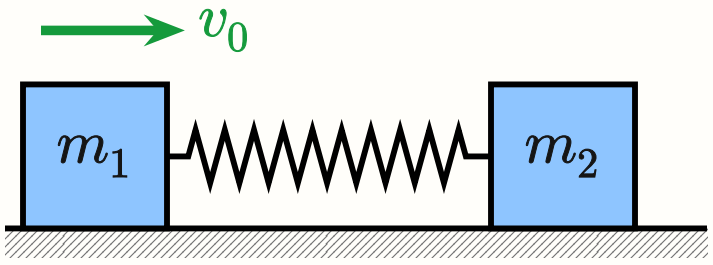
$$x_{max} = v\sqrt{\frac{m}{k}}$$

smooth floor, light spring · $\frac{1}{2}mv^2 = \frac{1}{2}kx_{max}^2$

Example

$m = 1\,\text{kg}$, $v = 4\,\text{m s}^{-1}$, $k = 400\,\text{N m}^{-1}$: $x_{max} = 0.2\,\text{m}$. *Rough floor,* $\mu_k = 0.5$: $8 = 200x^2 + 5x$, $x \approx 0.19\,\text{m}$.

2 Two blocks and a spring



$$\frac{1}{2}\mu v_0^2 = \frac{1}{2}kx_{max}^2$$

$$\mu = \frac{m_1 m_2}{m_1 + m_2}$$

μ: reduced mass · smooth floor (momentum conserved) · maximum compression at the SAME velocity

Example

$m_1 = m_2 = 1\,\text{kg}$, $v_0 = 2\,\text{m s}^{-1}$, $k = 100\,\text{N m}^{-1}$: *common velocity* $1\,\text{m s}^{-1}$, $x_{max} = 2\sqrt{0.5/100} \approx 0.141\,\text{m}$.

Golden points

★ *On a ROUGH floor a spring-pushed block is fastest where* $kx = \mu_k mg$ *(net force zero), not at the natural length.*

Vertical Spring Problems

1 Released at the natural length

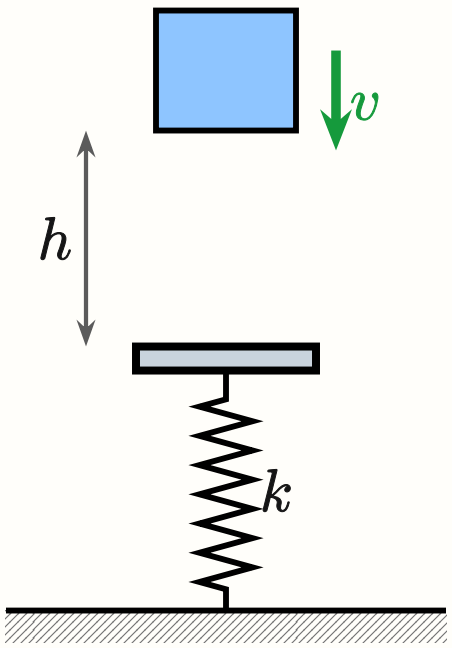
$$x_{eq} = \frac{mg}{k}, \qquad x_{max} = \frac{2mg}{k}$$

block on a hanging light spring, released from REST at the natural length

Example

$m = 2\,\text{kg}$, $k = 200\,\text{N m}^{-1}$: $x_{eq} = 0.1\,\text{m}$, $x_{max} = 0.2\,\text{m}$. *Lowered SLOWLY to* x_{eq}: *gravity gives* $2\,\text{J}$, *spring stores* $1\,\text{J}$, *hand takes* $1\,\text{J}$.

2 Block dropped on a spring



$$mg(h + x) = \tfrac{1}{2}kx^2$$

from rest, h *above the spring ·light spring ·* x*: maximum compression*

Example

$m = 1\,\text{kg}$, $h = 0.15\,\text{m}$, $k = 100\,\text{N m}^{-1}$: $50x^2 - 10x - 1.5 = 0$, $x = 0.3\,\text{m}$. *Fastest at* $x = mg/k = 0.1\,\text{m}$: $v^2 = 2(10)(0.25) - 100(0.1)^2/1 = 4$, $v_{max} = 2\,\text{m s}^{-1}$.

Golden points

★ *Fastest where net force* $= 0$ $(kx = mg)$; *maximum compression where* $v = 0$.

Power

1 Definitions

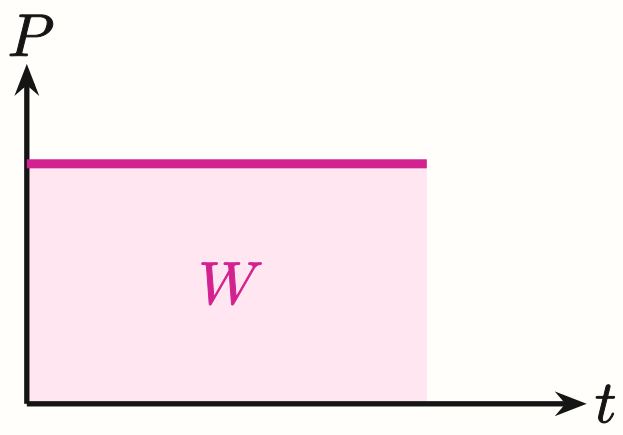
$$P_{avg} = \frac{W}{t}, \qquad P = \frac{dW}{dt} = \vec{F} \cdot \vec{v} = Fv\cos\theta$$

watt: $1\,\mathrm{W} = 1\,\mathrm{J\,s^{-1}}$, $[\mathrm{ML^2T^{-3}}]$ · $1\,\mathrm{hp} = 746\,\mathrm{W}$ · *scalar* · $\vec{F} \cdot \vec{v}$: *power of that force at that instant*

Example

Engine force $2000\,\mathrm{N}$ *at* $20\,\mathrm{m\,s^{-1}}$: $P = 40\,\mathrm{kW}$. $100\,\mathrm{kg}$ *raised at a steady* $2\,\mathrm{m\,s^{-1}}$: $P = mgv = 2000\,\mathrm{W}$.

2 Work from a P-t graph



- $W = \int P\,dt$ = *area under the* P-t *graph.*
- $1\,\mathrm{kWh} = 3.6 \times 10^6\,\mathrm{J}$ *(energy, not power).*

3 Constant power or constant force (from rest)

$$Pt = \tfrac{1}{2}mv^2 \;\Rightarrow\; v \propto t^{1/2},\; x \propto t^{3/2},\; F \propto t^{-1/2}$$

constant P, *all of it into* K *(no friction)* · *constant* F *instead:* $v \propto t$, $P = F^2t/m \propto t$

Example

$m = 2\,\mathrm{kg}$, $P = 4\,\mathrm{W}$ *from rest:* $v = \sqrt{2Pt/m} = 4\,\mathrm{m\,s^{-1}}$ *at* $t = 4\,\mathrm{s}$.

Efficiency, Pumps, Vehicles

1 Efficiency

$$\eta = P_{out}/P_{in} = W_{out}/W_{in} < 1$$

real machine: the rest is lost as heat

Example

Motor input $2\,\mathrm{kW}$, $\eta = 80\%$, *lifting* $100\,\mathrm{kg}$ *steadily:* $v = 1600/1000 = 1.6\,\mathrm{m\,s^{-1}}$.

2 Pump

$$P = \frac{dm}{dt}\left(gh + \tfrac{1}{2}v^2\right)$$

mass rate dm/dt *raised* h, *delivered at speed* v · *output power*

Example

$60\,\mathrm{kg}$ *of water a minute raised* $10\,\mathrm{m}$ *(slow delivery):* $100\,\mathrm{W}$; *at* $\eta = 50\%$, *input* $200\,\mathrm{W}$.

3 Vehicles

$$v_{max} = \frac{P}{f + mg\sin\theta}$$

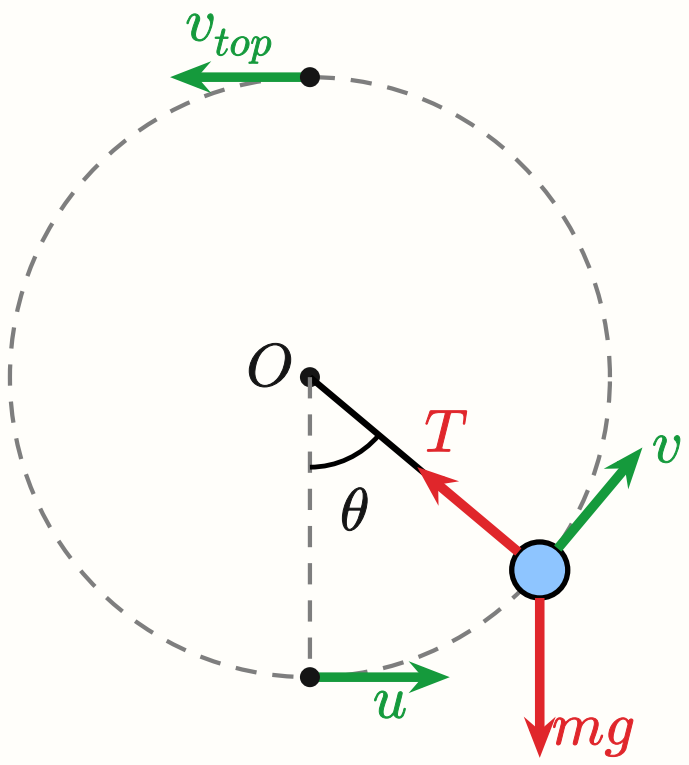
constant P, *steady speed (driving force = resistance* f *+ weight component)* · *level:* $\theta = 0$

Example

$P = 40\,\mathrm{kW}$, $f = 2000\,\mathrm{N}$, *level:* $v_{max} = 20\,\mathrm{m\,s^{-1}}$. *Same* $1000\,\mathrm{kg}$ *car,* $\sin\theta = 0.1$: $40000/3000 \approx 13.3\,\mathrm{m\,s^{-1}}$.

Vertical Circle: String

1 Bob on a string, given speed u at the bottom



$$v^2 = u^2 - 2gr(1 - \cos\theta)$$

$$T = \frac{mu^2}{r} - mg(2 - 3\cos\theta)$$

θ from the LOWEST point · light inextensible string of length r, no drag · valid while $T \geq 0$

- Radial equation: $T - mg\cos\theta = mv^2/r$; energy gives v. Tension does no work ($\perp \vec{v}$).

2 Special points

$$T_{low} = \frac{mu^2}{r} + mg, \qquad T_{hor} = \frac{mu^2}{r} - 2mg$$

$$T_{top} = \frac{mu^2}{r} - 5mg$$

$T_{low} - T_{top} = 6mg$ for any u that completes the circle

Example

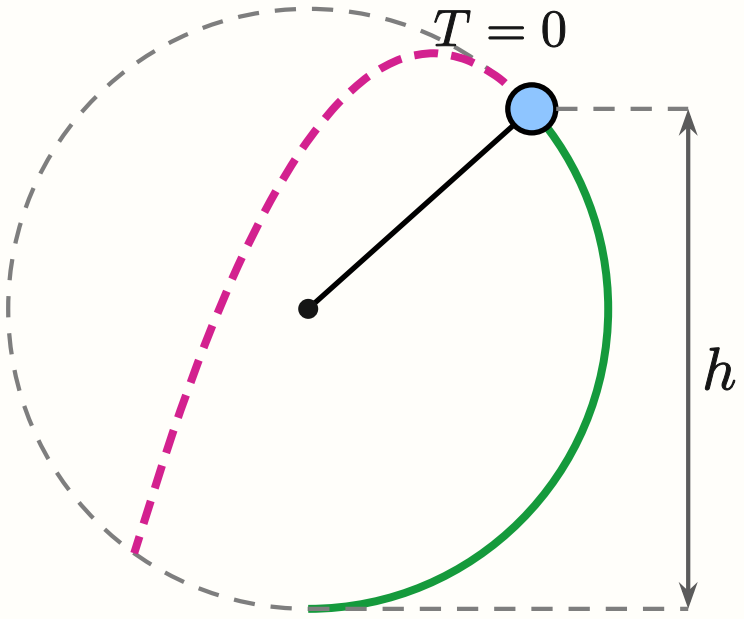
$m = 0.2\,\text{kg}$, $r = 0.5\,\text{m}$, $u = 5\,\text{m s}^{-1}$: $T_{low} = 10 + 2 = 12\,\text{N}$, $v_{top}^2 = 25 - 20 = 5$, $T_{top} = 2 - 2 = 0$: just completes.

Vertical Circle: Three Cases

1 Speed u at the bottom (string, length r)

$u \geq \sqrt{5gr}$	completes the circle ($T_{top} \geq 0$, $v_{top} \geq \sqrt{gr}$)	loops
$\sqrt{2gr} < u < \sqrt{5gr}$	string goes SLACK above the horizontal, then a parabola	leaves the circle
$u \leq \sqrt{2gr}$	stops at or below the horizontal, swings back; $T \geq 0$ throughout	oscillates

2 Leaving the circle



$$h = \frac{u^2 + gr}{3g}$$

$$v^2 = \frac{u^2 - 2gr}{3}$$

height above the LOWEST point where $T = 0$ · $\sqrt{2gr} < u < \sqrt{5gr}$ · v: speed there

Example

$r = 0.5\,\mathrm{m}$, $u = \sqrt{4gr} = \sqrt{20} \approx 4.47\,\mathrm{m\,s^{-1}}$: slack at $h = \frac{5}{3}r \approx 0.833\,\mathrm{m}$, $v = \sqrt{10/3} \approx 1.83\,\mathrm{m\,s^{-1}}$; then a projectile inside the circle (dashed).

Golden points

★ Speed is ZERO only in the oscillating case; when the string goes slack the bob is still moving ($v^2 = gr\sin\alpha$, α: angle of the string above the horizontal).

Rod, Track and Sphere

1 Minimum speed at the bottom (radius r)

light string; INSIDE a smooth vertical track (N for T): needs $v_{top} \geq \sqrt{gr}$

$$u_{min} = \sqrt{5gr}$$

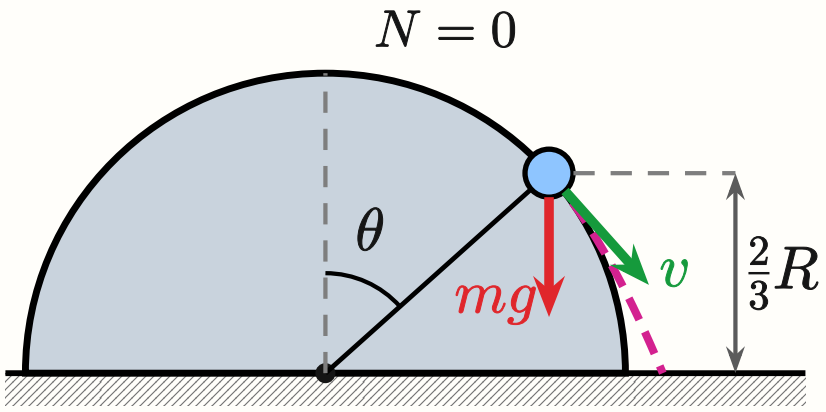
light rigid rod; bead on a ring; ball in a smooth tube (it can push): needs $v_{top} \geq 0$

$$u_{min} = \sqrt{4gr}$$

Example

Rod, $r = 0.5\,\mathrm{m}$: $u_{min} = \sqrt{20} \approx 4.47\,\mathrm{m\,s^{-1}}$. $v_{top} = 1\,\mathrm{m\,s^{-1}}$, $m = 0.2\,\mathrm{kg}$: $mv^2/r = 0.4\,\mathrm{N} < mg = 2\,\mathrm{N}$, *rod PUSHES up* $1.6\,\mathrm{N}$.

2 Outside a smooth sphere



$$\cos\theta = \tfrac{2}{3}$$

$$v = \sqrt{\tfrac{2}{3}gR}$$

from rest at the top, no friction · leaves where $N = 0$, $\frac{2}{3}R$ above the centre

Example

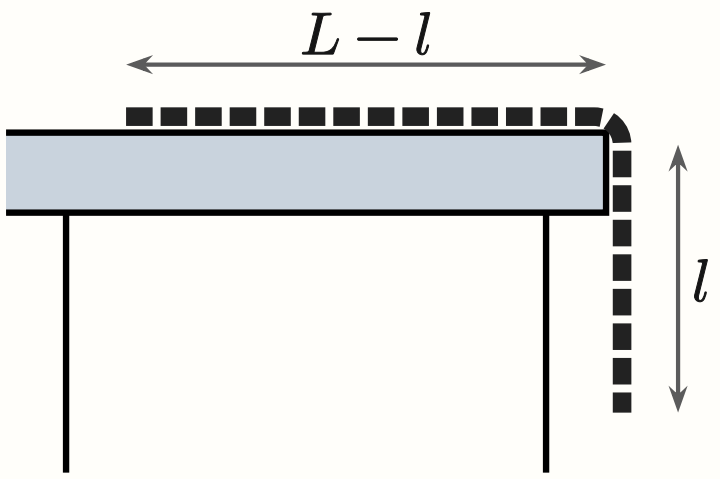
$R = 0.9\,\mathrm{m}$: *leaves* $0.3\,\mathrm{m}$ *below the top, at* $v = \sqrt{6} \approx 2.45\,\mathrm{m\,s^{-1}}$, *whatever its mass.*

Golden points

★ *Speed* u *at the top:* $\cos\theta = \dfrac{2 + u^2/gR}{3}$; *if* $u \geq \sqrt{gR}$ *it leaves at once.*

Chain on a Table

1 Pulling the hanging part back



$$W = \frac{mgL}{2n^2}$$

uniform chain m, L; $l = L/n$ hangs · smooth table, pulled slowly · hanging m/n rises $L/2n$

Example

$m = 3\,\mathrm{kg}$, $L = 1.2\,\mathrm{m}$, *one third hanging* ($n = 3$):

$$W = \frac{3 \times 10 \times 1.2}{2 \times 9} = 2\,\mathrm{J}.$$

2 Sliding off a smooth table

$$v = \sqrt{\frac{g\,(L^2 - l^2)}{L}}$$

from rest with l hanging · smooth table and edge · as the last link leaves

Example

$L = 1.2\,\mathrm{m}$, $l = 0.4\,\mathrm{m}$: $v = \sqrt{10 \times 1.28/1.2} \approx 3.27\,\mathrm{m\,s^{-1}}$.

Golden points

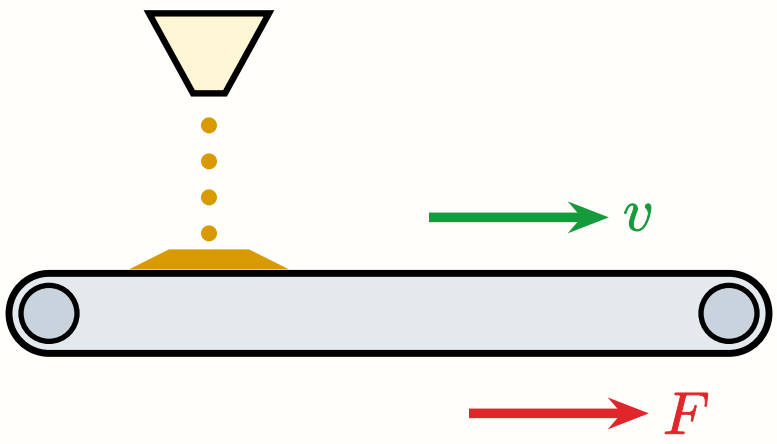
- ★ *Rough table: at rest if $l/L \le \mu_s/(1+\mu_s)$; $\mu_s = 0.25$: up to 20% hangs.*
- ★ *Kinetic friction while it slides off: $W_f = -\mu_k mg(L-l)^2/2L$.*

Ropes and Variable Mass

1 Lifting a rope or a leaky bucket

- *Uniform rope lying on the floor, lifted slowly by one end until it just hangs: its centre of mass rises $L/2$, $W = mgL/2$.* $4\,\mathrm{kg}$, $2\,\mathrm{m}$: $40\,\mathrm{J}$.
- *Bucket raised slowly, mass falling uniformly from* $20\,\mathrm{kg}$ *to* $10\,\mathrm{kg}$ *over* $10\,\mathrm{m}$ *($\lambda = 1\,\mathrm{kg\,m^{-1}}$):*
  $W = \int_0^{10} (20 - y) g\,dy = 10(200 - 50) = 1500\,\mathrm{J}$.

2 Conveyor belt



$$F = \frac{dm}{dt}\,v, \qquad P = \frac{dm}{dt}\,v^2$$

sand falls at rate dm/dt on a belt kept at v · F: extra pull needed · half to K, half heat

Example

$2\,\mathrm{kg\,s^{-1}}$ *onto a belt at* $3\,\mathrm{m\,s^{-1}}$: $F = 6\,\mathrm{N}$, $P = 18\,\mathrm{W}$: $9\,\mathrm{W}$ *into* K, $9\,\mathrm{W}$ *heat.*

3 Chain falling on a scale pan

$$N = 3\lambda g x$$

uniform chain (λ per length) released from rest hanging just above the pan, links falling freely · x: length already on the pan · $= 3\times$ its weight

- *Weight on the pan $\lambda g x$ plus the rate of momentum delivered, $\lambda v^2 = \lambda(2gx)$.*

Forms of Energy and Units

1 Forms of energy

- Mechanical ($K+U$), thermal (internal), chemical, electrical, nuclear, sound, light. Energy changes FORM; the total of an isolated system stays constant.
- Mass is energy too: $E=mc^2$; $1\,\mathrm{g}$ of mass is equivalent to $9\times10^{13}\,\mathrm{J}$.

2 Units of energy

$1\,\mathrm{erg}=10^{-7}\,\mathrm{J}$ $\qquad$ $1\,\mathrm{eV}=1.6\times10^{-19}\,\mathrm{J}$

$1\,\mathrm{cal}=4.186\,\mathrm{J}$ $\qquad$ $1\,\mathrm{kWh}=3.6\times10^{6}\,\mathrm{J}$

3 Link: collisions (separate chapter)

$$\sum\vec{p}_i=\sum\vec{p}_f$$

no external impulse during the collision · KINETIC energy is conserved ONLY in a perfectly elastic collision ($e=1$) · perfectly inelastic: $e=0$, bodies stick

- In an inelastic collision the lost K becomes heat, sound and deformation: TOTAL energy is still conserved.
- Full treatment (1D, 2D, coefficient of restitution) in the Centre of Mass and Collisions notes.

Golden points

★ Explosion (internal forces only): momentum conserved, K INCREASES (chemical energy released).

Common Traps

1 Exam traps

Avoid these

- ✘ Work is a vector (it is a scalar; it can be negative).
- ✘ Friction always does negative work (static friction can do positive).
- ✘ N never does work (only on a FIXED surface).
- ✘ Spring $10 \to 20\,\text{cm}$ = $0 \to 10\,\text{cm}$ (it is 3 times).
- ✘ Mechanical energy always conserved (only if $W_{nc} = 0$).
- ✘ Internal forces cannot change K (explosion does).
- ✘ Every equilibrium is stable (a maximum of U is unstable: check U'').
- ✘ String: $u_{min} = \sqrt{4gr}$ (rod; string needs $\sqrt{5gr}$).
- ✘ Bob stops where the string goes slack (it still moves).

2 Solving recipe

Steps

- List the forces; mark which do work (in which frame).
- Only conservative work: conserve $K + U$; else $W_{nc} = \Delta E_{mech}$.
- Circle: energy for v, radial $F = mv^2/r$ for T or N.

Summary

$$W = \int \vec{F} \cdot d\vec{r} \;\bullet\; W_{net} = \Delta K \;\bullet\; F = -\frac{dU}{dx} \;\bullet\; P = \vec{F} \cdot \vec{v}$$

Questions: JEE Main Level (I)

$g = 10\,\mathrm{m\,s^{-2}}$ *throughout*

Q1 · JEE Main level · MCQ

$\vec{F} = (3\hat{i} + 4\hat{j})\,\mathrm{N}$ moves a particle from $(1\,\mathrm{m}, 2\,\mathrm{m})$ to $(4\,\mathrm{m}, -2\,\mathrm{m})$. Work done: (A) $7\,\mathrm{J}$ (B) $-7\,\mathrm{J}$ (C) $25\,\mathrm{J}$ (D) 0

Solution: $\vec{d} = (3\hat{i} - 4\hat{j})\,\mathrm{m}$, $W = 9 - 16 = -7\,\mathrm{J}$.

Answer: (B)

Q2 · JEE Main level · Numerical

$F = (4x + 3)\,\mathrm{N}$ acts along x. Work (in J) from $x = 1\,\mathrm{m}$ to $x = 3\,\mathrm{m}$?

Solution: $W = \left[2x^2 + 3x\right]_1^3 = 27 - 5 = 22\,\mathrm{J}$.

Answer: 22

Q3 · JEE Main level · MCQ

The kinetic energy of a body increases by 300%. Its momentum increases by: (A) 50% (B) 100% (C) 200% (D) 300%

Solution: $K' = 4K$, $p \propto \sqrt{K}$: $p' = 2p$, an increase of 100%.

Answer: (B)

Q4 · JEE Main level · Numerical

A $0.5\,\mathrm{kg}$ bob on a $1\,\mathrm{m}$ string just completes a vertical circle. Tension (in N) at the lowest point?

Solution: $u^2 = 5gr$: $T = mu^2/r + mg = 6mg = 30\,\mathrm{N}$.

Answer: 30

Questions: JEE Main Level (2)

$g = 10\,\mathrm{m\,s^{-2}}$ *throughout*

Q5 · JEE Main level · Numerical

A pump of efficiency 50% *lifts* $300\,\mathrm{kg}$ *of water per minute from a well* $20\,\mathrm{m}$ *deep (delivered slowly). Input power in kW?*

Solution: $P_{out} = \dfrac{300 \times 10 \times 20}{60} = 1000\,\mathrm{W}$; $P_{in} = 1000/0.5 = 2\,\mathrm{kW}$.

Answer: 2

Q6 · JEE Main level · Numerical

A $2\,\mathrm{kg}$ *block moving at* $3\,\mathrm{m\,s^{-1}}$ *on a smooth floor hits a light spring of* $k = 200\,\mathrm{N\,m^{-1}}$. *Maximum compression in cm?*

Solution: $x = v\sqrt{m/k} = 3\sqrt{0.01} = 0.3\,\mathrm{m}$.

Answer: 30

Q7 · JEE Main level · MCQ

$U(x) = (2x^2 - 8x + 5)\,\mathrm{J}$. *The particle is in equilibrium at: (A)* $x = 2\,\mathrm{m}$, *stable (B)* $x = 2\,\mathrm{m}$, *unstable (C)* $x = 4\,\mathrm{m}$, *stable (D) nowhere*

Solution: $F = -(4x - 8) = 0$ *at* $x = 2\,\mathrm{m}$; $U'' = 4 > 0$: *stable.*

Answer: (A)

Q8 · JEE Main level · Numerical

A $1\,\mathrm{kg}$ *block slides from rest down a rough incline of* 37° *and length* $5\,\mathrm{m}$ ($\mu_k = 0.5$). *Kinetic energy (in J) at the bottom?*

Solution: $W_g = 1 \times 10 \times 3 = 30\,\mathrm{J}$,
$W_f = -0.5 \times 1 \times 10 \times 0.8 \times 5 = -20\,\mathrm{J}$, $K = 10\,\mathrm{J}$.

Answer: 10

Questions: JEE Advanced Level (I)

$g = 10\,\mathrm{m\,s^{-2}}$ *throughout*

A1 · JEE Advanced level · One or more correct

A $1\,\mathrm{kg}$ particle moves along x in $U(x) = (x^4 - 2x^2)\,\mathrm{J}$ (x in m), released from rest at $x = 0.5\,\mathrm{m}$. Correct: (A) $x = 0$ is stable (B) $x = \pm 1\,\mathrm{m}$ are stable (C) it moves between $x = 0.5\,\mathrm{m}$ and $x \approx 1.32\,\mathrm{m}$ (D) its maximum speed is about $1.06\,\mathrm{m\,s^{-1}}$

Solution: $U' = 4x(x^2 - 1)$; $U'' = 12x^2 - 4$: -4 at 0 (unstable), 8 at ± 1 (stable). $E = U(0.5) = -0.4375\,\mathrm{J}$; $U = E$ gives $x^2 = 1 \pm 0.75$: $x = 0.5$, $\sqrt{1.75} \approx 1.32\,\mathrm{m}$ ($U(0) = 0 > E$, so it cannot cross 0). At $x = 1$: $K = 1 - 0.4375 = 0.5625\,\mathrm{J}$, $v \approx 1.06\,\mathrm{m\,s^{-1}}$.

Answer: (B), (C), (D)

A2 · JEE Advanced level · Numerical

A bob on a light string of length $0.9\,\mathrm{m}$ is given $u = \sqrt{4gL} = 6\,\mathrm{m\,s^{-1}}$ at the lowest point. Find the greatest height (m) above the lowest point it reaches.

Solution: Slack at $h = \dfrac{u^2 + gL}{3g} = \dfrac{36 + 9}{30} = 1.5\,\mathrm{m}$ ($\sin\alpha = \frac{2}{3}$ above the centre) with $v^2 = \dfrac{36 - 18}{3} = 6$. Vertical part of v: $v\cos\alpha$, $v^2\cos^2\alpha = 6 \times \frac{5}{9} = \frac{10}{3}$; it rises $\dfrac{10/3}{20} \approx 0.167\,\mathrm{m}$ more: $H \approx 1.67\,\mathrm{m}$ ($< 2L$).

Answer: ≈ 1.67

Questions: JEE Advanced Level (2)

$g = 10\,\mathrm{m\,s^{-2}}$ *throughout*

A3 · JEE Advanced level · Paragraph

A $2\,\mathrm{kg}$ block is pressed against a light spring ($k = 800\,\mathrm{N\,m^{-1}}$) on a rough floor ($\mu_k = 0.2$), compressing it $0.1\,\mathrm{m}$, and released from rest (not attached). (i) speed as the spring reaches its natural length; (ii) maximum speed; (iii) total distance moved.

Solution: $f = \mu_k mg = 4\,\mathrm{N}$, $\frac{1}{2}kx_0^2 = 4\,\mathrm{J}$. (i) $K = 4 - 4(0.1) = 3.6\,\mathrm{J}$, $v = \sqrt{3.6} \approx 1.90\,\mathrm{m\,s^{-1}}$. (ii) fastest at $kx = f$, $x = 0.005\,\mathrm{m}$: $K = 400(0.01 - 0.000025) - 4(0.095) = 3.61\,\mathrm{J}$, $v_{max} = 1.9\,\mathrm{m\,s^{-1}}$. (iii) after leaving: $3.6/4 = 0.9\,\mathrm{m}$; total $1.0\,\mathrm{m}$.

Answer: (i) $\approx 1.90\,\mathrm{m\,s^{-1}}$ (ii) $1.9\,\mathrm{m\,s^{-1}}$ (iii) $1.0\,\mathrm{m}$

A4 · JEE Advanced level · Matching

Bob on a string of length L, speed u at the lowest point. Match. (P) $u = \sqrt{2gL}$ (Q) $u = \sqrt{3gL}$ (R) $u = \sqrt{5gL}$ (S) $u = \sqrt{6gL}$ with (1) just reaches the horizontal and stops (2) string slack at $\frac{4}{3}L$ above the bottom (3) $T_{top} = 0$ (4) $T_{top} = mg$

Solution: (P) $v = 0$ at $h = u^2/2g = L$. (Q) $h = \dfrac{3gL + gL}{3g} = \frac{4}{3}L$. (R) $T_{top} = 5mg - 5mg = 0$. (S) $T_{top} = 6mg - 5mg = mg$.

Answer: P-1, Q-2, R-3, S-4

Tip

★ Vertical circle: compare u^2 with $2gL$ and $5gL$ FIRST to know which case you are in.

Questions: JEE Advanced Level (3)

$g = 10\,\mathrm{m\,s^{-2}}$ *throughout*

A5 · JEE Advanced level · One or more correct

A small block slides from rest at the top of a smooth fixed hemisphere of radius $R = 0.9\,\mathrm{m}$ resting on the ground. Correct: (A) it leaves the surface $0.6\,\mathrm{m}$ above the ground (B) its speed then is $\sqrt{6}\,\mathrm{m\,s^{-1}}$ (C) the height depends on its mass (D) the normal force is zero there

Solution: $N = 0$: $mg\cos\theta = mv^2/R$ with $v^2 = 2gR(1-\cos\theta)$: $\cos\theta = \tfrac{2}{3}$, height $\tfrac{2}{3}R = 0.6\,\mathrm{m}$, $v^2 = \tfrac{2}{3}gR = 6$. Mass cancels.

Answer: (A), (B), (D)

A6 · JEE Advanced level · One or more correct

A machine delivers constant power $P = 4\,\mathrm{W}$ to a $2\,\mathrm{kg}$ body starting from rest on a smooth floor. Correct: (A) $v \propto t^{1/2}$ (B) $x \propto t^{3/2}$ (C) the force on it $\propto t^{-1/2}$ (D) $v = 4\,\mathrm{m\,s^{-1}}$ and $x \approx 10.7\,\mathrm{m}$ at $t = 4\,\mathrm{s}$

Solution: $Pt = \tfrac{1}{2}mv^2$: $v = \sqrt{2Pt/m} = 2\sqrt{t}$; $x = \tfrac{4}{3}t^{3/2}$; $F = P/v = 2/\sqrt{t}$. At $t = 4\,\mathrm{s}$: $v = 4\,\mathrm{m\,s^{-1}}$, $x = \tfrac{4}{3}(8) \approx 10.7\,\mathrm{m}$.

Answer: (A), (B), (C), (D)

How to use these

- ★ Attempt each question before reading its solution; then redo the ones you missed after a day.
- ★ Every answer here can be checked by a second method (energy and force, or a limiting case).